System Overview and Architecture

Status date: April 29, 2026 **Audience:** the next developer who picks up Omnilore Scheduler from Team 34 **Prerequisites:** familiarity with Flutter/Dart and basic AWS EC2 concepts

1. Problem and Audience

Omnilore is a member-led lifelong-learning organization in Rancho Palos Verdes, CA. Every term, members rank course preferences (first choices and backups), declare their availability across twenty time slots, and request a specific number of classes (0–6). A small administrative team then has to compute a term schedule that:

1. drops courses that fall under a minimum class size,
2. splits oversize courses into two or more sections,
3. assigns each surviving course to one of twenty time slots,
4. assigns one main coordinator and a co-coordinator (or two equal co-coordinators) to each course, and
5. exposes four export actions: an early roster in phone-roster style, a final roster with coordinator labels, a mail-merge file, and an unmet-wants report.

The previous (pre-2026) implementation was a Flutter desktop application that depended on dart:io, native window sizing, and OS file dialogs. Onboarding a new admin meant installing Flutter and the desktop runtime. Several stakeholder-reported defects (the "Co-Coordinator" assignment bug and the "Show Splits" bug) and a missing intermediate-state save/load capability remained outstanding.

Team 34's mandate (CSCI 401, Spring 2026) was to migrate the application to a web-based delivery model, fix the two defects, add intermediate state save/load aligned with admin expectations, expand regression testing, and stand up automated cloud deployment. An optional AI-assisted scheduling feature was scoped out by the stakeholder in favor of stability.

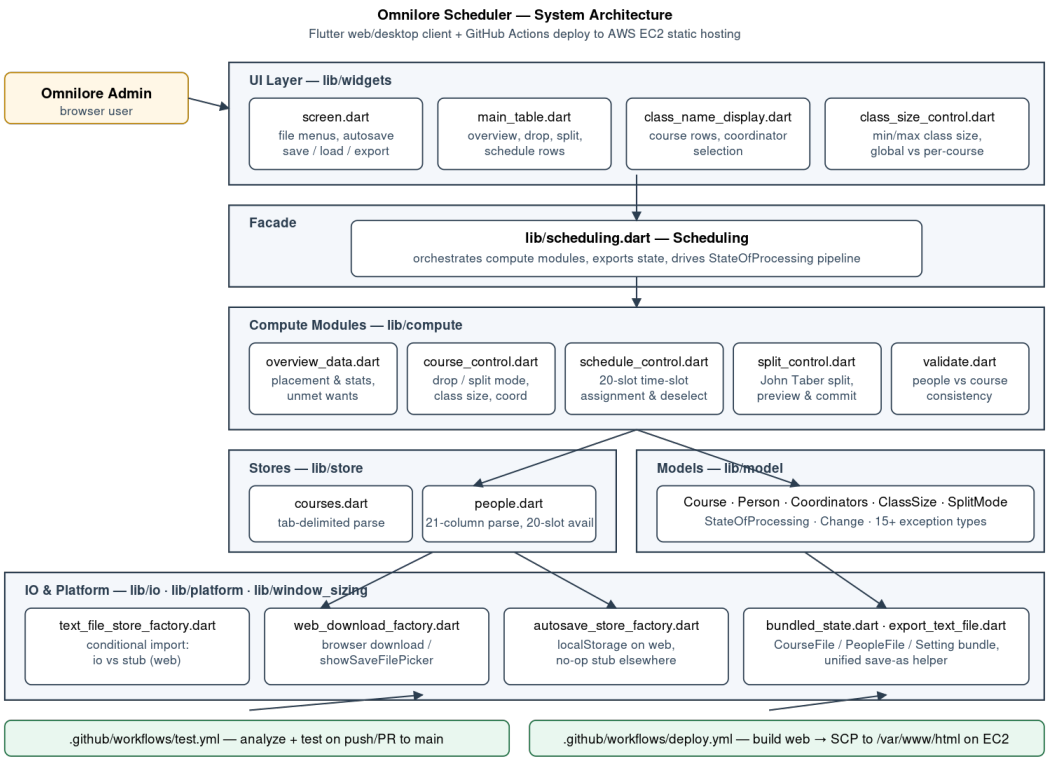
2. Features Delivered (D1 → D7)

- **Browser-deployable build.** The same Dart codebase compiles for desktop and web. Conditional imports route file I/O, downloads, autosave, and window sizing to platform-appropriate implementations. A static check (scripts/check_platform_gating.sh) prevents dart:io and other desktop-only imports from leaking into shared code.
- **Two-file load.** Tab-delimited course and people text files are parsed in-browser via file_picker and validated against the thirteen exception types defined in lib/model/exceptions.dart.
- **Drop/Split/Schedule/Coordinator pipeline.** The original desktop pipeline is preserved end-to-end; oversize courses now expose a non-destructive **Show Splits preview** with manual rebalancing and explicit Cancel / Implement controls.

- **Coordinator UX rebuild.** Both Set C and CC and Set CC1 and CC2 now reset selection state on re-entry so admins can fully reassign coordinators; tapping a highlighted name on a coordinator-assigned course clears the assignment.
- **Time-slot deselection.** Clicking the same slot a second time unschedules the course; downstream availability counts recompute.
- **Save / Save As / Load / Autosave / Restore.** lib/io/bundled_state.dart produces a single text bundle containing the original course file, the original people file, and the scheduler state grammar — restoring everything from one file. Web autosave uses six localStorage keys with timestamped restore-prompts.
- **Four export menu actions.** Early roster (phone-roster style), final roster (with (C) / (CC) labels), tab-delimited mail-merge, and unmet-wants report. All exports route through a unified helper (lib/io/export_text_file.dart) with timestamped filenames from lib/io/default_filename.dart; web exports use showSaveFilePicker where the browser supports it and fall back to a normal browser download.
- **Cloud deployment.** .github/workflows/deploy.yml builds the Flutter web target, briefly whitelists the runner IP in an EC2 security group, copies build/web/* to /var/www/html over SCP, and revokes the IP — reachable today at http://scheduler.omnilore.org.
- **Continuous validation.** .github/workflows/test.yml runs the gating check, flutter analyze, and flutter test on every push and PR to main.

3. Architecture

The architecture splits cleanly into a UI layer, a single Scheduling facade, the pure-Dart compute and model modules behind it, conditionally-imported I/O abstractions, and a CI/CD path that builds the web target and pushes it to EC2.



Layer responsibilities

Layer	Path	Job
UI	lib/widgets/	Owns Flutter state and renders the main table; hands user actions to Scheduling and reacts to its callback. The most active widget files are screen.dart, class_name_display.dart, class_size_control.dart, and widgets/table/*.
Facade	lib/scheduling.dart	The only object the UI talks to. Loads files, delegates compute to the modules below, computes StateOfProcessing after every change, and produces all exports. ~809 LOC.
Compute	lib/compute/	Pure scheduling logic (overview_data.dart, course_control.dart, schedule_control.dart,

		split_control.dart, validate.dart). No Flutter, no dart:io.
Models	lib/model/	Plain-Dart data carriers: Course, Person, Coordinators, ClassSize, SplitMode, Change, StateOfProcessing, exceptions.dart.
Stores	lib/store/	Tab-delimited parsers for the course and people files. Throw the exception types in lib/model/exceptions.dart on malformed input.
IO	lib/io/, lib/platform/, lib/window_sizing/	Conditional-import factories that pick a desktop or browser implementation at compile time.

Conditional-import pattern (platform gating)

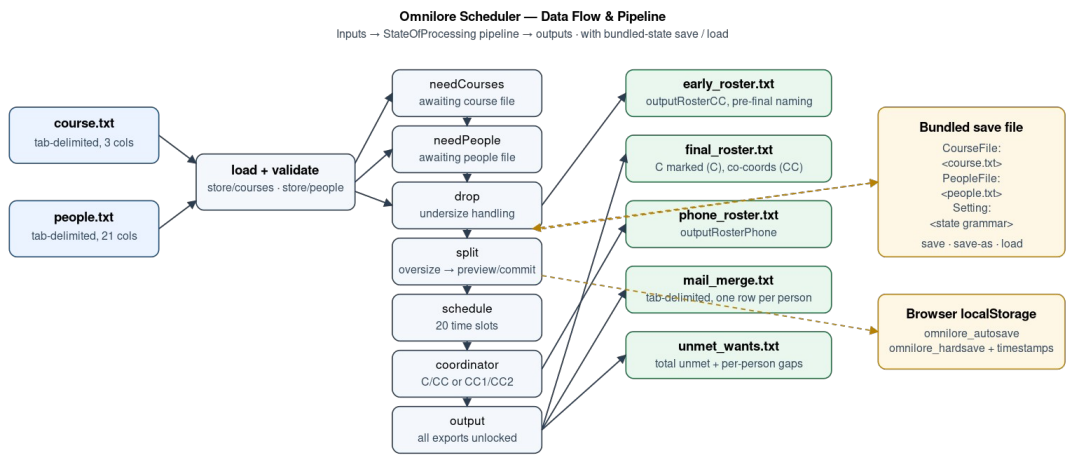
Every desktop-only API is hidden behind a three-file factory:

```
// text_file_store_factory.dart
export 'text_file_store_stub.dart'
    if (dart.library.io) 'package:omnilore_scheduler/io/text_file_store_io.dart';
```

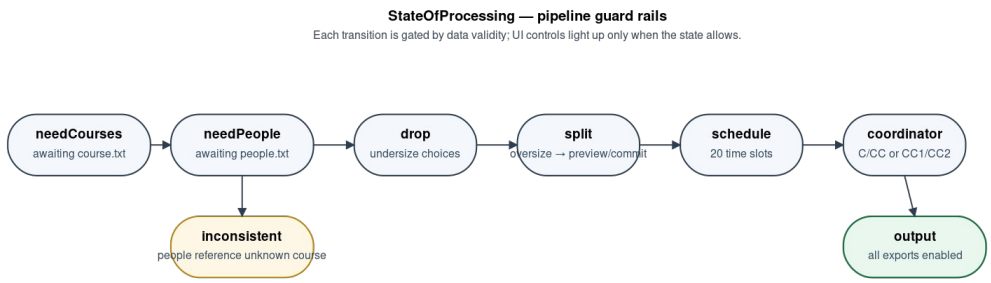
scripts/check_platform_gating.sh greps the entire lib/ tree and **fails the build** if import 'dart:io'; or import 'package:desktop_window/…'; appears anywhere outside the explicit allowlist. This is also enforced by test/platform_gating_test.dart. Whenever you add a new platform-touching feature, follow the same triple of *_factory.dart + *_stub.dart + *_io.dart (or *_web.dart) and update the gating script.

4. Data Flow

Two tab-delimited input files (3-column course offerings and 21-column member preferences) feed the Scheduling facade, which walks each course through the drop / split / schedule / coordinator pipeline and into one of four timestamped exports. The same facade also produces the bundled save (course file plus people file plus state grammar) that powers Save As, Load, and the browser autosave.



5. Pipeline State Machine



Scheduling._updateStateOfProcessing() evaluates the eight StateOfProcessing values **in order** after every UI-driven compute(Change) call:

needCourses → needPeople → inconsistent | drop | split | schedule | coordinator → output

State	Entered when	UI status label (stateDescriptions in screen.dart)	What the UI gates on next
needCourses	course store is empty	"Need Courses"	Admin uses **File → Import Course**.
needPeople	course OK, people store is empty	"Need People"	Admin uses **File → Import People**.
inconsistent	a person references an unknown course code	"Inconsistent"	Admin must fix the input file (offending code is in the exception message).
drop	at least one course is below its minimum class size	"Drop and Split"	Drop / split-mode controls light up on the affected rows.

split	at least one course is over its maximum size	"Drop and Split"	Show Splits preview is required before commit.
schedule	all classes sized; some unscheduled	"Schedule"	The twenty-slot table is live; click-to-assign / click-again-to-deselect.
coordinator	all scheduled; coordinators incomplete	"Coordinator"	Set C and CC and Set CC1 and CC2 controls light up.
output	everything assigned	"Output"	All four exports are unlocked.

The pipeline is **monotonic forward** but reactive backward: dropping a course or rebalancing a split rewinds downstream stages and the UI greys out their controls until the upstream constraint is satisfied again.

6. Key Design Choices

- **Single Dart engine for desktop and web.** The compute and model layers are pure Dart with no Flutter or dart:io dependencies. This keeps the desktop-vs-web fork to the I/O surface only, which is the smallest possible blast radius.
- **Bundled save format.** A single text file (CourseFile: + PeopleFile: + Setting: sections) restores the entire working session. This was chosen over JSON because (a) it remains diff-friendly and human-readable for stakeholder review, (b) it allows admins to hand-edit a snapshot if needed, and (c) it preserves the original tab-delimited input bytes — useful for traceability and re-running.
- **Autosave is convenience, not durability.** Web autosave writes to browser localStorage and is therefore tied to that browser profile and device. The user-facing message and the runbook treat Save As to a real file as the only operational backup.
- **No backend.** The application is entirely client-side. There is no database, no auth, no API, and no PII server-side. PII (names, phone numbers) lives only in the admin's browser session and in the files they choose to save.
- **Static-asset deploy.** The web build is just static files. EC2 + nginx/Apache was the path of least resistance because Omnilore already has an AWS account; the doc includes a recommendation to migrate to S3 + CloudFront in the long term (see Backlog_Known_Issues_Roadmap.md, P2).
- **Aggressive platform gating.** Catching a desktop-only import at CI time is cheap; debugging a runtime dart:io failure in a stakeholder's browser is not. The grep-based gating script and matching test prevent that.

7. Known Limits

- The app is a single-user, client-side scheduling tool. There is no shared state, multi-user collaboration, audit log, or user identity.

- Browser autosave is browser-profile-scoped. Switching browsers, profiles, or devices loses the autosave; admins must Save As to a portable bundle.
- showSaveFilePicker support varies by browser. On unsupported browsers (notably Firefox at the time of writing) the save flow falls back to a normal download.
- AWS deployment currently overwrites /var/www/html on EC2 in place. There is no automated rollback artifact; rollback is "redeploy a known-good commit" (see Operations_Runbook.md).
- Performance under unusually large datasets (well beyond the 24-course / 267-person fixture) is not separately profiled.
- The optional AI-assisted scheduling enhancement from D1 was deferred by stakeholder agreement and is **not** delivered.

8. Where to extend the system

Need	Touch this	Watch out for
Add a new export	New outputXyzToString() on Scheduling; route through exportTextFile()	Keep the gate check on StateOfProcessing so disabled actions stay disabled.
Add a new pipeline stage	New StateOfProcessing enum value + branch in _updateStateOfProcessing(); new compute module under lib/compute/	Update the bundled-state grammar and the legacy state loader (Scheduling.loadState).
Add CSV exports	Wrap the existing string-builders with a CSV serializer alongside the tab-delimited form	Mail-merge already produces tab-delimited rows; adding a CSV output is mostly a header + escaping pass.
Move off EC2 to S3 + CloudFront	Replace deploy.yml with aws s3 sync build/web s3://… + an invalidation step	Update Operations_Runbook.md and the rotation table; archive the EC2 host config.
Add a backend (e.g., shared term schedules)	A new lib/api/ package with conditional Web/desktop transport; gate it the same way as I/O	Introduces auth and PII at rest — re-evaluate Security_Privacy_Accessibility_UX_Notes.md before implementing.

9. Repository inventory (for the next developer)

```
lib/
  scheduling.dart              # facade — start here
  main.dart                    # Flutter entry + theme
  widgets/                     # UI
    screen.dart                # main screen, file menus, autosave
    class_name_display.dart    # course rows + coordinator selection
```

```
    class_size_control.dart       # min/max class-size controls
    overview_data.dart            # overview/stats panel
    table/{main_table, drop_row, schedule_row, class_name_row,
overview_row}.dart
    names_display_mode.dart, utils.dart
  compute/
    overview_data.dart            # placement + unmet wants
    course_control.dart           # drop, split mode, class size, coordinators
    schedule_control.dart         # 20-slot scheduling state
    split_control.dart            # split preview/commit (John Taber's algorithm)
    validate.dart                 # people vs courses consistency
  model/
    course.dart, person.dart, coordinators.dart, class_size.dart
    state_of_processing.dart, change.dart, exceptions.dart
  store/
    courses.dart, people.dart, list_item.dart
  io/
    text_file_store_factory.dart   # conditional import gate
    text_file_store_stub.dart, text_file_store_io.dart, text_file_store.dart
    web_download_factory.dart, web_download.dart, web_download_stub.dart,
web_download_web.dart
    autosave_store_factory.dart, autosave_store_stub.dart, autosave_store_web.dart
    bundled_state.dart            # Save / Save As / Load bundle
    default_filename.dart         # timestamped export/save filenames
    export_text_file.dart         # unified save-as helper
    file_io_exceptions.dart
  platform/
    desktop_platform.dart, desktop_platform_stub.dart, desktop_platform_io.dart
  window_sizing/
    window_sizing.dart, window_sizing_desktop.dart, window_sizing_noop.dart
  theme.dart
test/                             # 21 .dart test files, 90 passing tests (see
Testing_and_QA_Summary.md)
.github/workflows/
  test.yml                        # CI
  deploy.yml                      # build web → SCP to EC2
scripts/
  check_platform_gating.sh        # grep-based desktop-import gate
docs/                             # earlier-deliverable working docs
Project_Docs_D7/                  # D7 handoff source docs + rendered deliverables
```